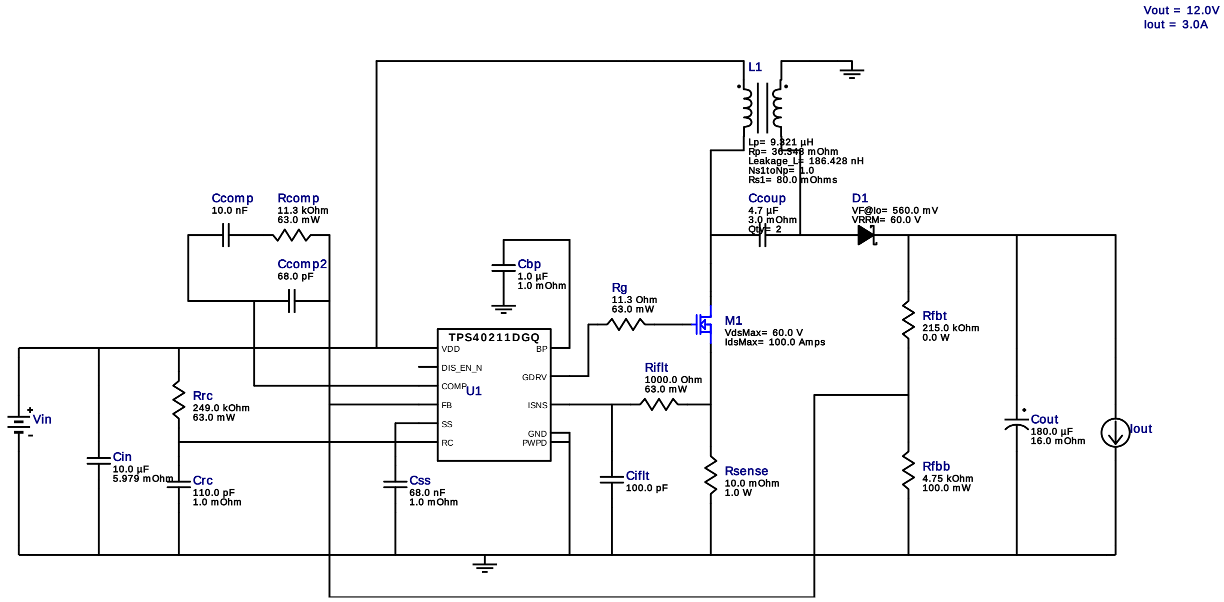
VinMin = 12.0V	Device = TPS40211DGQR
VinMax = 16.8V	Topology = SEPIC
Vout = 12.0V	Created = 2026-05-31 18:17:32.167
Iout = 3.0A	BOM Cost = NA
	BOM Count = 21
	Total Pd = 3.45W

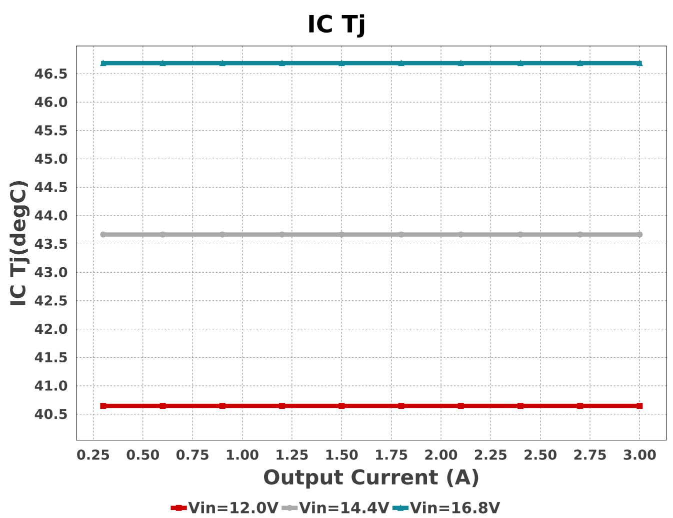
Design : 5 TPS40211DGQR
TPS40211DGQR 12V-16.8V to 12.00V @ 3A

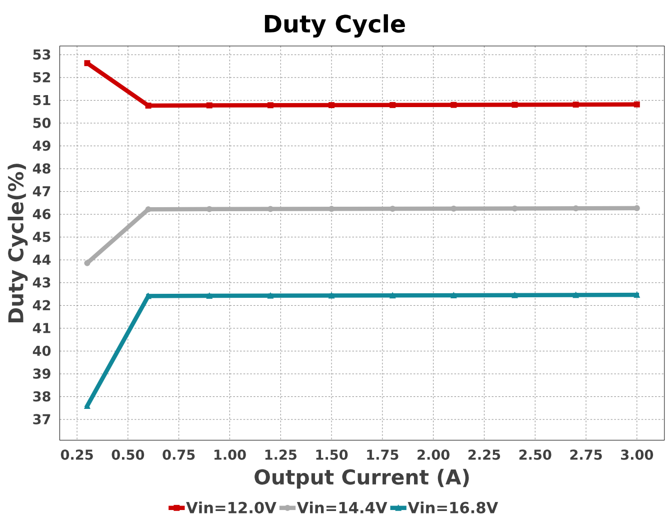


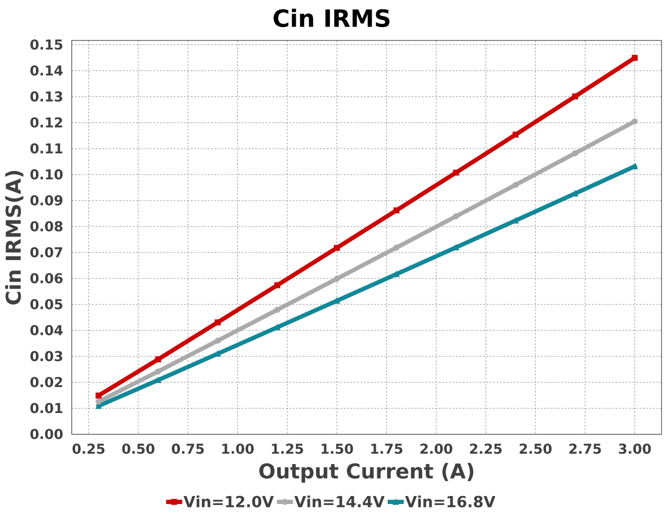
Electrical BOM

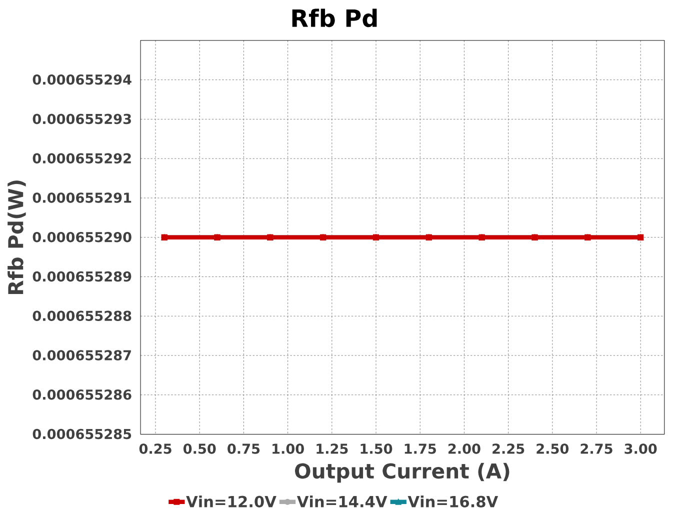
Name	Manufacturer	Part Number	Properties	Qty	Price	Footprint
Cbp	Taiyo Yuden	EMK107B7105KA-T Series= X7R	Cap= 1.0 uF ESR= 1.0 mOhm VDC= 16.0 V IRMS= 0.0 A	1	$0.01	0603 5 mm²
Ccomp	TDK	C2012C0G1H103K060AA Series= C0G/NP0	Cap= 10.0 nF VDC= 50.0 V IRMS= 0.0 A	1	$0.07	0805 7 mm²
Ccomp2	Yageo	CC0805JRNPO9BN680 Series= C0G/NP0	Cap= 68.0 pF VDC= 50.0 V IRMS= 0.0 A	1	$0.01	0805 7 mm²
Ccoup	MuRata	GRM31CR71H475KA12L Series= X7R	Cap= 4.7 uF ESR= 3.0 mOhm VDC= 50.0 V IRMS= 4.98 A	2	$0.10	1206 11 mm²
Ciflt	Samsung Electro-Mechanics	CL21C101JBANNNC Series= C0G/NP0	Cap= 100.0 pF VDC= 50.0 V IRMS= 0.0 A	1	$0.01	0805 7 mm²
Cin	Taiyo Yuden	MSASG21GBB5106MTNA01 Series= X5R	Cap= 10.0 uF ESR= 5.979 mOhm VDC= 35.0 V IRMS= 1.88442 A	1	$0.17	0805 7 mm²
Cout	Panasonic	25SVPF180M Series= SVPF	Cap= 180.0 uF ESR= 16.0 mOhm VDC= 25.0 V IRMS= 4.65 A	1	$1.17	CAPSMT_62_E12 106 mm²
Crc	MuRata	GRM1555C1E111JA01D Series= C0G/NP0	Cap= 110.0 pF ESR= 1.0 mOhm VDC= 25.0 V IRMS= 0.0 A	1	$0.01	0402 3 mm²

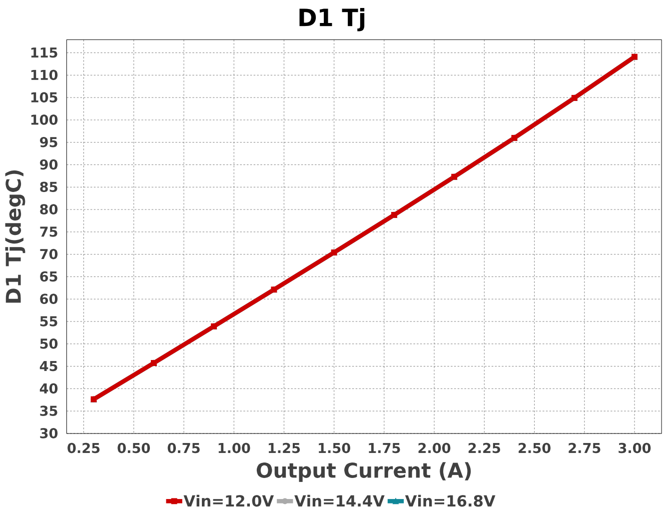
Name	Manufacturer	Part Number	Properties	Qty	Price	Footprint
Css	MuRata	GRM155R61A683KA01D Series= X5R	Cap= 68.0 nF ESR= 1.0 mOhm VDC= 10.0 V IRMS= 0.0 A	1	$0.01	0402 3 mm²
D1	Diodes Inc.	PDS760-13	VF@Io= 560.0 mV VRRM= 60.0 V	1	$0.36	PowerDI5 50 mm²
L1	CUSTOM	CUSTOM	Lp= 9.321 µH Rp= 36.348 mOhm Leakage_L= 186.428 nH Ns1toNp= 1.0 Rs1= 80.0 mOhms	1	NA	CUSTOM 0 mm²
M1	Texas Instruments	CSD18534KCS	VdsMax= 60.0 V IdsMax= 100.0 Amps	1	$0.38	KCS0003B 80 mm²
Rcomp	Vishay-Dale	CRCW040211K3FKED Series= CRCW..e3	Res= 11.3 kOhm Power= 63.0 mW Tolerance= 1.0%	1	$0.01	0402 3 mm²
Rfbb	Susumu Co Ltd	RR1220P-4751-D-M Series= RR12	Res= 4.75 kOhm Power= 100.0 mW Tolerance= 0.5%	1	$0.02	0805 7 mm²
Rfbt	CUSTOM	CUSTOM Series= ?	Res= 215.0 kOhm Power= 0.0 W Tolerance= 0.0%	1	NA	CUSTOM 0 mm²
Rg	Vishay-Dale	CRCW040211R3FKED Series= CRCW..e3	Res= 11.3 Ohm Power= 63.0 mW Tolerance= 1.0%	1	$0.01	0402 3 mm²
Riflt	Vishay-Dale	CRCW04021K00FKED Series= CRCW..e3	Res= 1000.0 Ohm Power= 63.0 mW Tolerance= 1.0%	1	$0.01	0402 3 mm²
Rrc	Yageo	RC0402FR-07249KL Series= ?	Res= 249.0 kOhm Power= 63.0 mW Tolerance= 1.0%	1	$0.01	0402 3 mm²
Rsense	Susumu Co Ltd	PRL1632-R010-F-T1 Series= PRL1632	Res= 10.0 mOhm Power= 1.0 W Tolerance= 1.0%	1	$0.20	0612 11 mm²
U1	Texas Instruments	TPS40211DGQR	Switcher	1	$0.72	DGQ0010D_NV_N 24 mm²

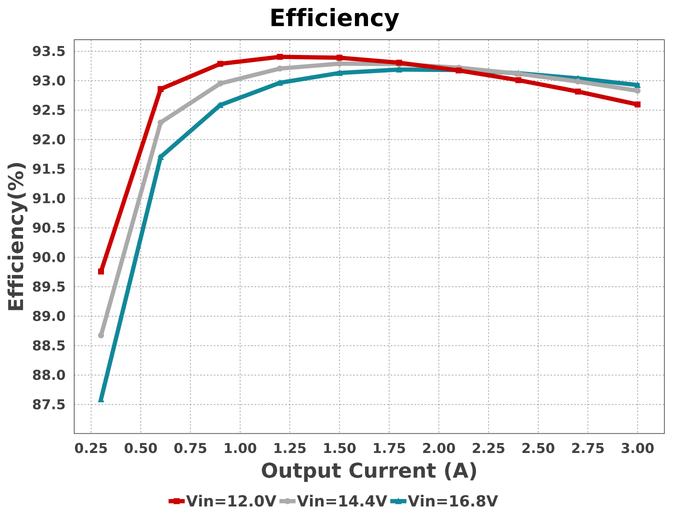


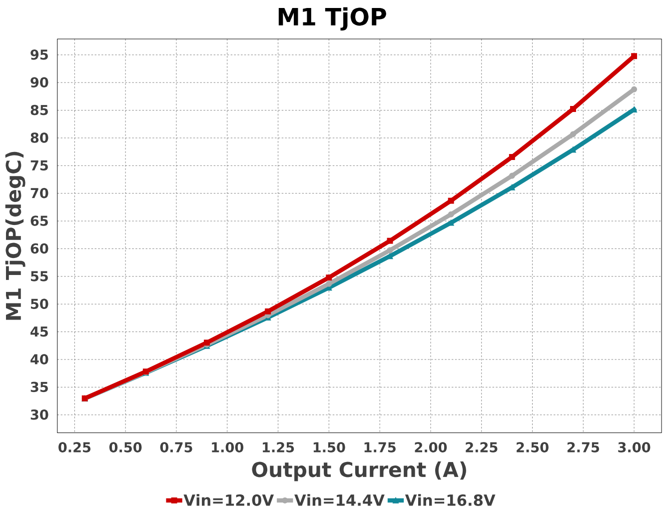


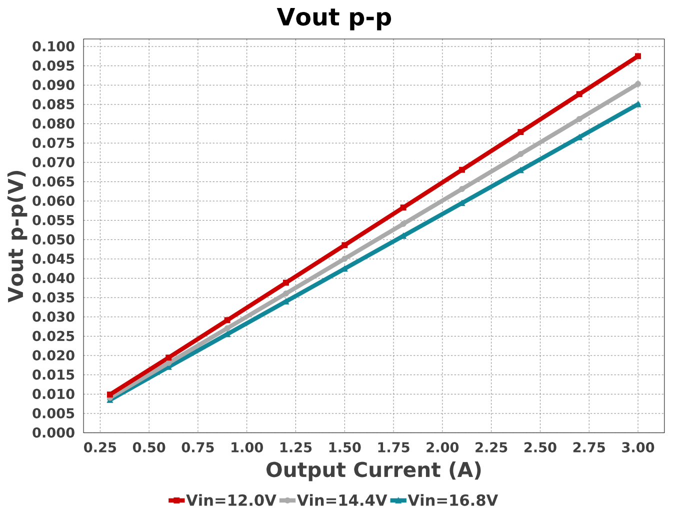

Cin IRMS
Cin IRMS(A)
0.15
0.14
0.13
0.12
0.11
0.10
0.09
0.08
0.07
0.06
0.05
0.04
0.03
0.02
0.01
0.00
0.25 0.50 0.75 1.00 1.25 1.50 1.75 2.00 2.25 2.50 2.75 3.00
Output Current (A)
Vin=12.0V Vin=14.4V Vin=16.8V

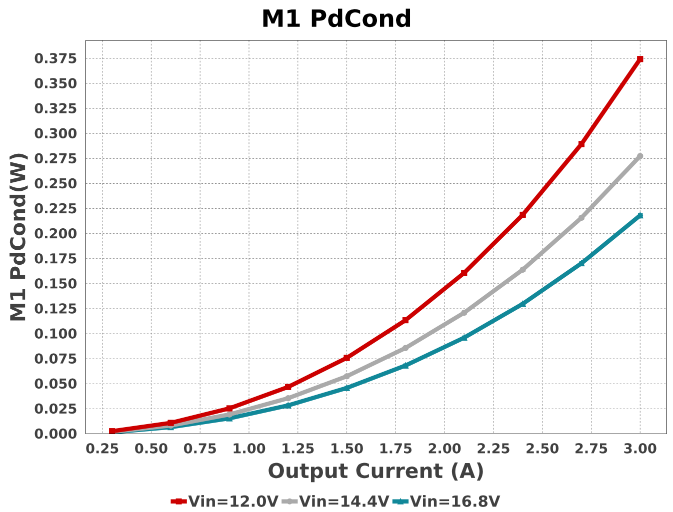

Rfb Pd
Rfb Pd(W)
0.000655294
0.000655293
0.000655292
0.000655291
0.000655290
0.000655289
0.000655288
0.000655287
0.000655286
0.000655285
0.25 0.50 0.75 1.00 1.25 1.50 1.75 2.00 2.25 2.50 2.75 3.00
Output Current (A)
Vin=12.0V Vin=14.4V Vin=16.8V

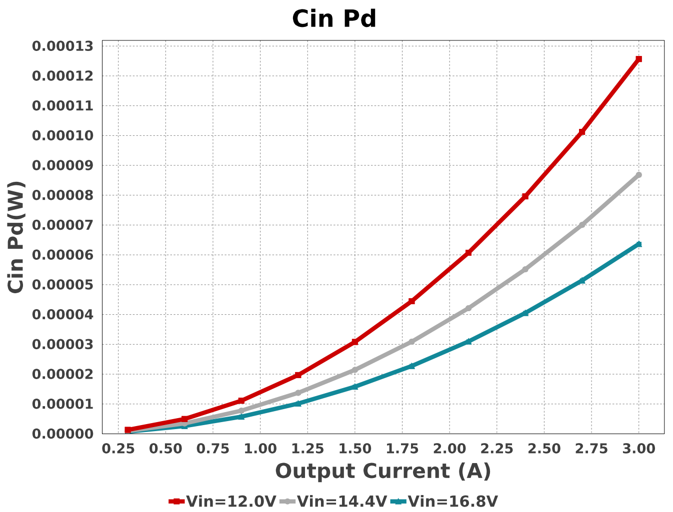

D1 Tj
D1 Tj(degC)
115
110
105
100
95
90
85
80
75
70
65
60
55
50
45
40
35
30
0.25 0.50 0.75 1.00 1.25 1.50 1.75 2.00 2.25 2.50 2.75 3.00
Output Current (A)
Vin=12.0V Vin=14.4V Vin=16.8V

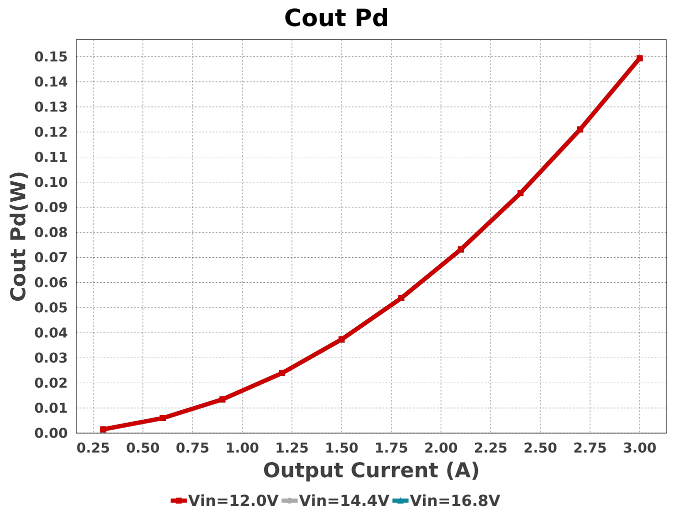

Efficiency
Efficiency(%)
93.5
93.0
92.5
92.0
91.5
91.0
90.5
90.0
89.5
89.0
88.5
88.0
87.5
0.25 0.50 0.75 1.00 1.25 1.50 1.75 2.00 2.25 2.50 2.75 3.00
Output Current (A)
Vin=12.0V Vin=14.4V Vin=16.8V

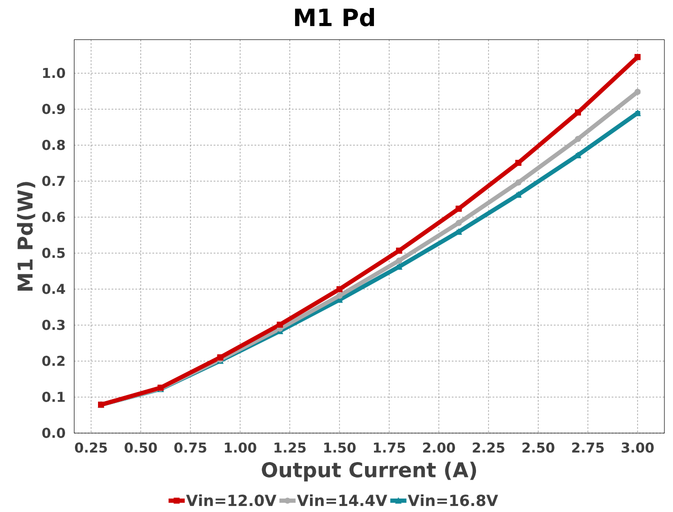

M1 TjOP
M1 TjOP(degC)
95
90
85
80
75
70
65
60
55
50
45
40
35
30
0.25 0.50 0.75 1.00 1.25 1.50 1.75 2.00 2.25 2.50 2.75 3.00
Output Current (A)
Vin=12.0V Vin=14.4V Vin=16.8V

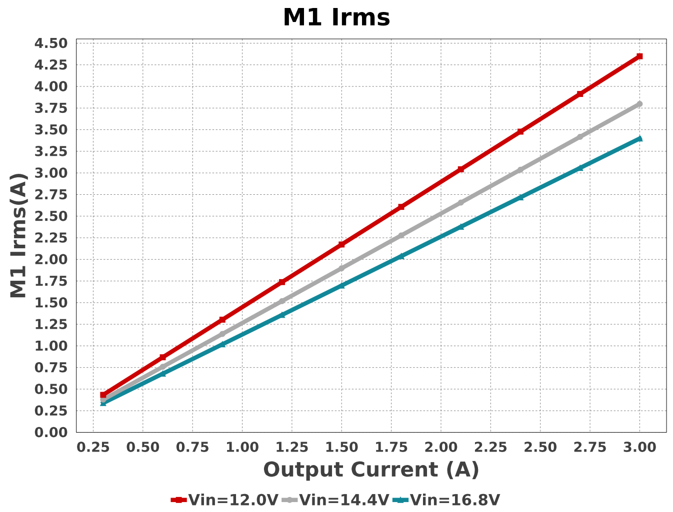

Vout p-p
Vout p-p(V)
0.100
0.095
0.090
0.085
0.080
0.075
0.070
0.065
0.060
0.055
0.050
0.045
0.040
0.035
0.030
0.025
0.020
0.015
0.010
0.005
0.000
0.25 0.50 0.75 1.00 1.25 1.50 1.75 2.00 2.25 2.50 2.75 3.00
Output Current (A)
Vin=12.0V Vin=14.4V Vin=16.8V

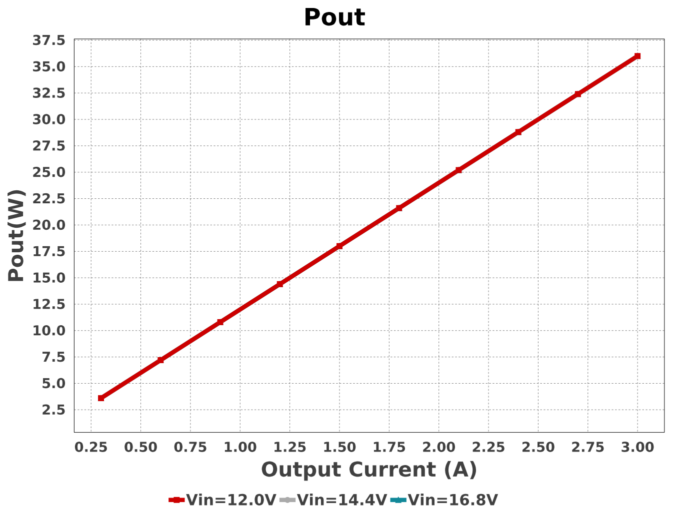

M1 PdCond
M1 PdCond(W)
Output Current (A)
Vin=12.0V Vin=14.4V Vin=16.8V


Cin Pd
Cin Pd(W)
Output Current (A)
Vin=12.0V Vin=14.4V Vin=16.8V

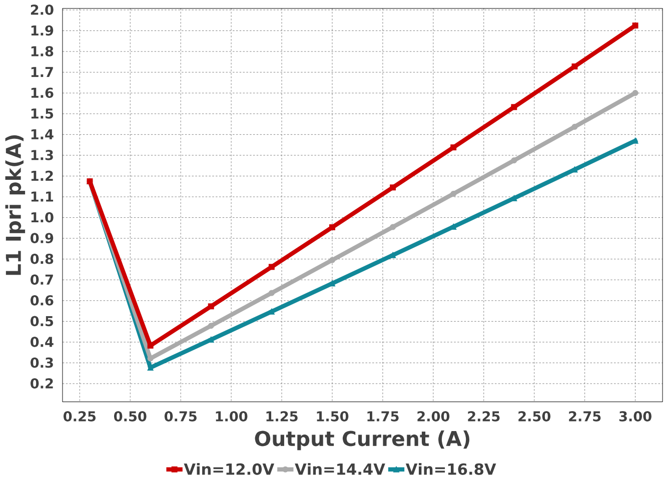

Cout Pd
Cout Pd(W)
Output Current (A)
Vin=12.0V Vin=14.4V Vin=16.8V


M1 Pd
M1 Pd(W)
Output Current (A)
Vin=12.0V Vin=14.4V Vin=16.8V

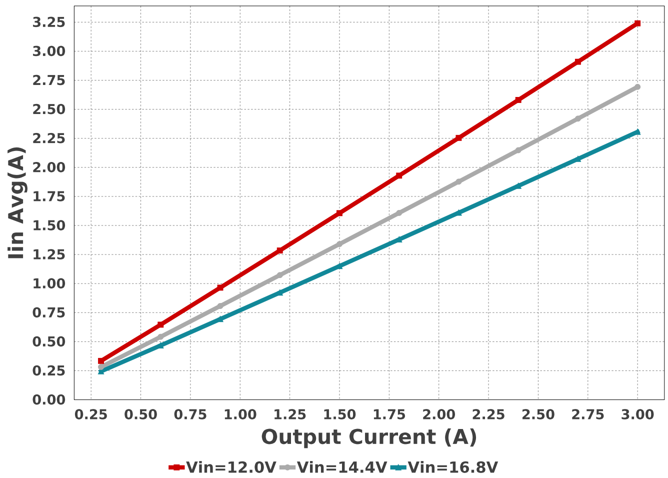

M1 Irms
M1 Irms(A)
Output Current (A)
Vin=12.0V Vin=14.4V Vin=16.8V


Pout
Pout(W)
Output Current (A)
Vin=12.0V Vin=14.4V Vin=16.8V

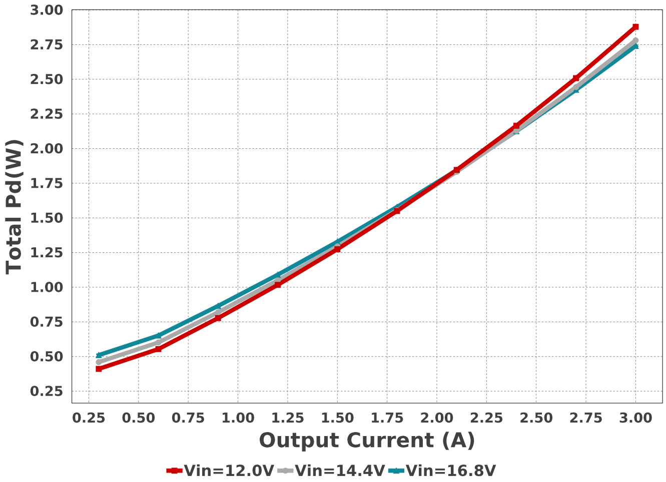
L1 Ipri pk



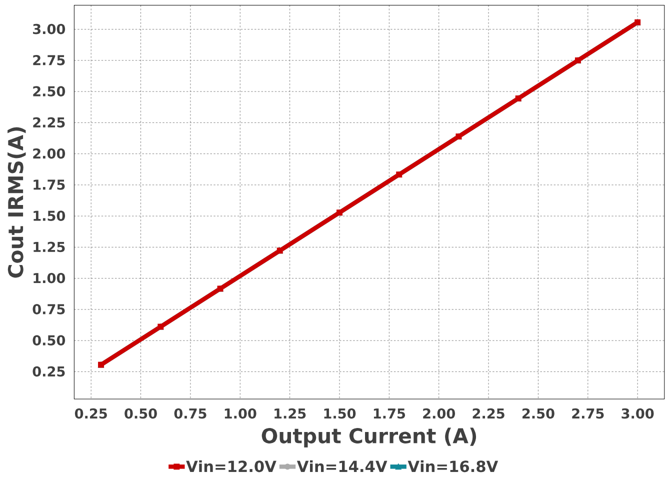
Iin Avg



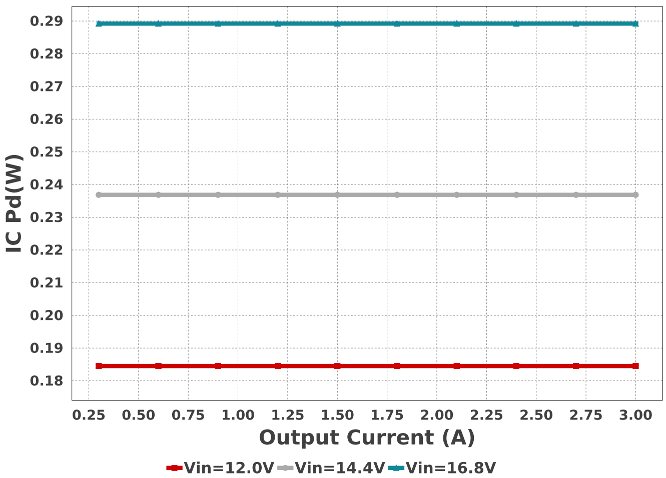
Total Pd



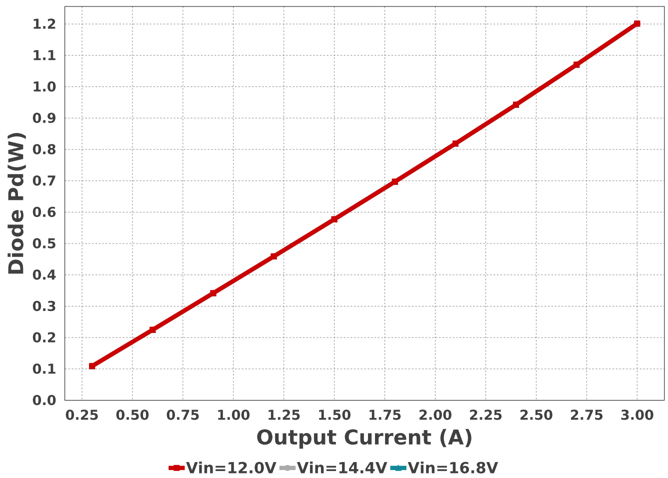
Cout IRMS

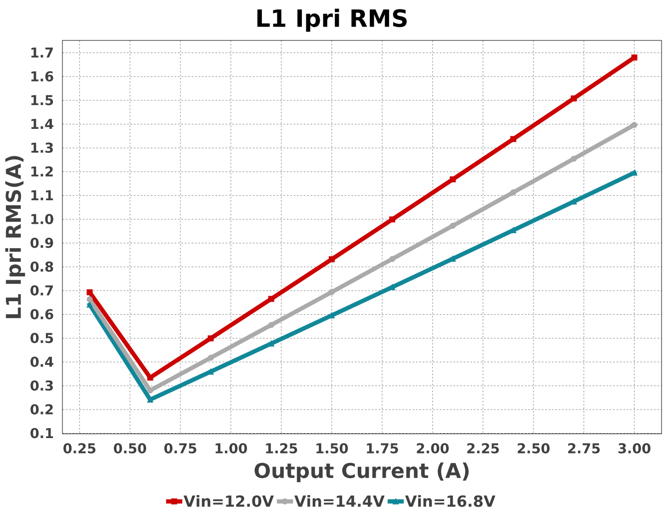


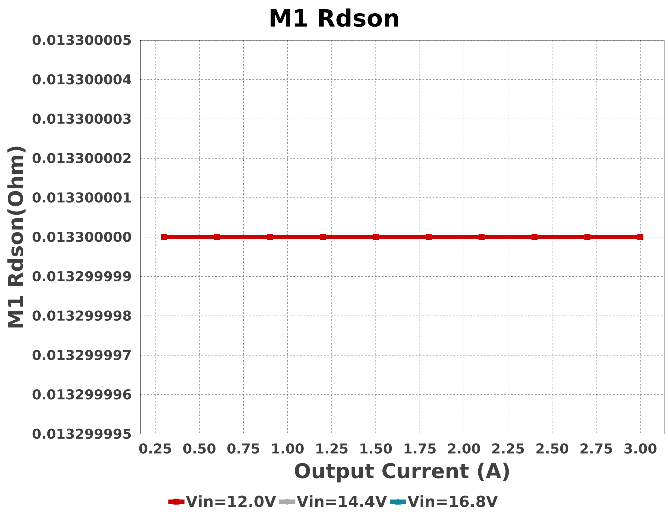
IC Pd

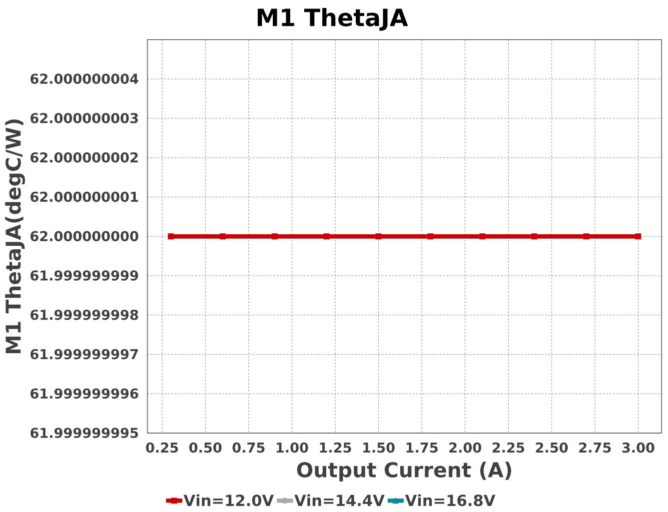


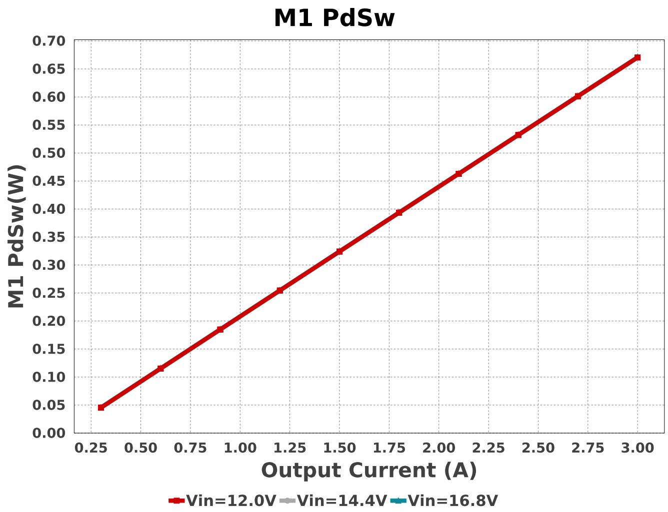
Diode Pd

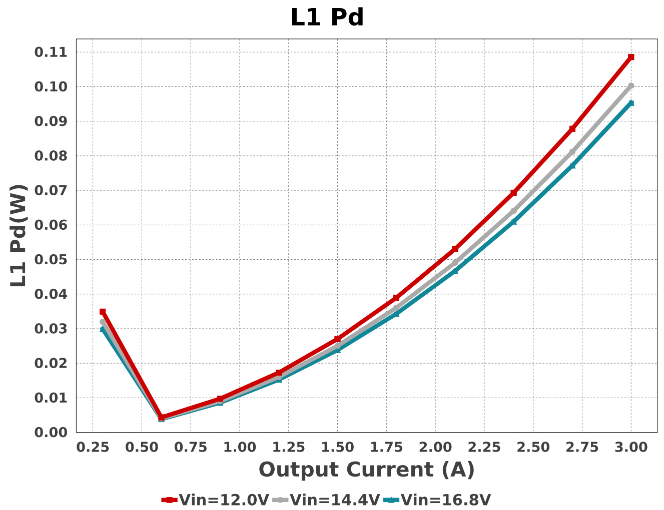


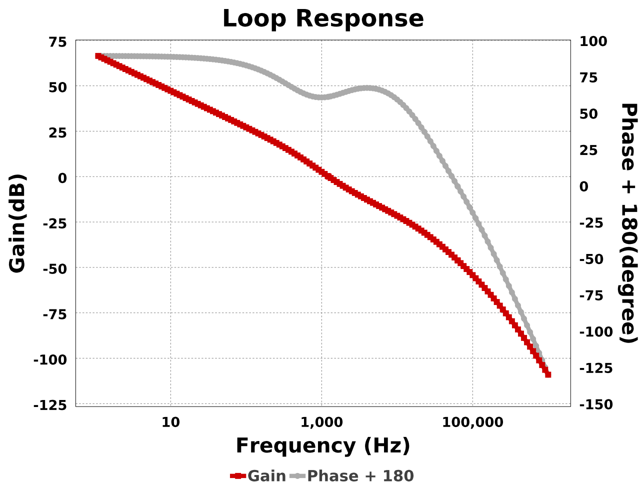












Operating Values

#	Name	Value	Category	Description
1.	Cin IRMS	147.15 mA	Capacitor	Input capacitor RMS ripple current
2.	Cin Pd	129.46 µW	Capacitor	Input capacitor power dissipation
3.	Cout IRMS	3.056 A	Capacitor	Output capacitor RMS ripple current
4.	Cout Pd	149.4 mW	Capacitor	Output capacitor power dissipation
5.	D1 Tj	115.2 degC	Diode	D1 junction temperature
6.	Diode Pd	1.217 W	Diode	Diode power dissipation
7.	IC Pd	173.03 mW	IC	IC power dissipation
8.	IC Tj	39.984 degC	IC	IC junction temperature
9.	IC Tolerance	14.0 mV	IC	IC Feedback Tolerance
10.	ICThetaJA	57.7 degC/W	IC	IC junction-to-ambient thermal resistance
11.	Iin Avg	3.287 A	IC	Average input current

#	Name	Value	Category	Description
12.	M1 Irms	4.35 A	Mosfet	M1 MOSFET Irms
13.	M1 Pd	1.609 W	Mosfet	M1 MOSFET total power dissipation
14.	M1 PdCond	436.23 mW	Mosfet	M1 MOSFET conduction losses
15.	M1 PdSw	1.173 W	Mosfet	M1 MOSFET switching losses
16.	M1 Rdson	13.3 mOhm	Mosfet	Drain-Source On-resistance
17.	M1 ThetaJA	62.0 degC/W	Mosfet	MOSFET junction-to-ambient thermal resistance
18.	M1 TjOP	129.75 degC	Mosfet	M1 MOSFET junction temperature
19.	Cin Pd	129.46 µW	Power	Input capacitor power dissipation
20.	Cout Pd	149.4 mW	Power	Output capacitor power dissipation
21.	Diode Pd	1.217 W	Power	Diode power dissipation
22.	IC Pd	173.03 mW	Power	IC power dissipation
23.	L1 Pd	109.38 mW	Power	Power Dissipation in the Inductor
24.	M1 Pd	1.609 W	Power	M1 MOSFET total power dissipation
25.	M1 PdCond	436.23 mW	Power	M1 MOSFET conduction losses
26.	M1 PdSw	1.173 W	Power	M1 MOSFET switching losses
27.	Rfb Pd	655.29 µW	Power	Rfb Power Dissipation
28.	Total Pd	3.448 W	Power	Total Power Dissipation
29.	Rfb Pd	655.29 µW	Resistor	Rfb Power Dissipation
30.	BOM Count	21	System Information	Total Design BOM count
31.	Cross Freq	1.26 kHz	System Information	Bode plot crossover frequency
32.	Duty Cycle	50.831 %	System Information	Duty cycle
33.	Efficiency	91.26 %	System Information	Steady state efficiency
34.	FootPrint	553.0 mm^2	System Information	Total Foot Print Area of BOM components
35.	Frequency	534.018 kHz	System Information	Switching frequency
36.	Gain Marg	-46.301 dB	System Information	Bode Plot Gain Margin
37.	Iout	3.0 A	System Information	Iout operating point
38.	Low Freq Gain	68.119 dB	System Information	Gain at 1Hz
39.	Mode	CCM	System Information	Conduction Mode
40.	Phase Marg	59.094 deg	System Information	Bode Plot Phase Margin
41.	Pout	36.0 W	System Information	Total output power
42.	Total BOM	NA	System Information	Total BOM Cost
43.	Vin	12.0 V	System Information	Vin operating point
44.	Vout	12.0 V	System Information	Operational Output Voltage
45.	Vout Actual	12.028 V	System Information	Vout Actual calculated based on selected voltage divider resistors
46.	Vout Tolerance	5.903 %	System Information	Vout Tolerance based on IC Tolerance (no load) and voltage divider resistors if applicable
47.	Vout p-p	99.207 mV	System Information	Peak-to-peak output ripple voltage
48.	L1 Ipri RMS	1.706 A	Transformer	RMS current in primary of PFC-Coil L1.
49.	L1 Ipri pk	1.954 A	Transformer	Peak current in primary of PFC-Coil L1.
50.	L1 Pd	109.38 mW	Transformer	Power Dissipation in the Inductor

Design Inputs

Name	Value	Description
Iout	3.0	Maximum Output Current
VinMax	16.8	Maximum input voltage
VinMin	12.0	Minimum input voltage
VinTyp	14.8	Typical input voltage
Vout	12.0	Output Voltage
base_pn	TPS40211	Base Product Number
source	DC	Input Source Type
Ta	30.0	Ambient temperature

WEBENCH® Assembly

Component Testing

Some published data on components in datasheets such as Capacitor ESR and Inductor DC resistance is based on conservative values that will guarantee that the components always exceed the specification. For design purposes it is usually better to work with typical values. Since this data is not always available it is a good practice to measure the Capacitance and ESR values of Cin and Cout, and the inductance and DC resistance of L1 before assembly of the board. Any large discrepancies in values should be electrically simulated in WEBENCH to check for instabilities and thermally simulated in WebTHERM to make sure critical temperatures are not exceeded.

Soldering Component to Board

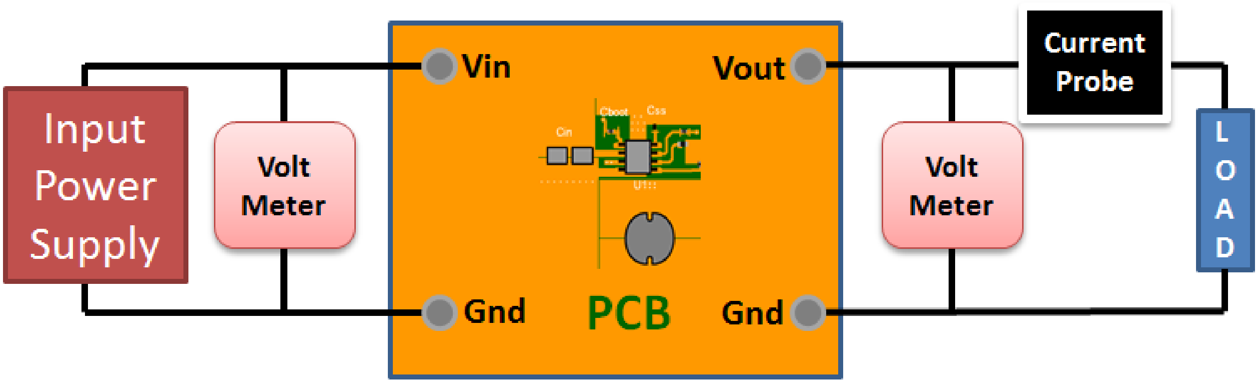
If board assembly is done in house it is best to tack down one terminal of a component on the board then solder the other terminal. For surface mount parts with large tabs, such as the DPAK, the tab on the back of the package should be pre-tinned with solder, then tacked into place by one of the pins. To solder the tab town to the board place the iron down on the board while resting against the tab, heating both surfaces simultaneously. Apply light pressure to the top of the plastic case until the solder flows around the part and the part is flush with the PCB. If the solder is not flowing around the board you may need a higher wattage iron (generally 25W to 30W is enough).

Initial Startup of Circuit

It is best to initially power up the board by setting the input supply voltage to the lowest operating input voltage 12.0V and set the input supply's current limit to zero. With the input supply off connect up the input supply to Vin and GND. Connect a digital volt meter and a load if needed to set the minimum Iout of the design from Vout and GND. Turn on the input supply and slowly turn up the current limit on the input supply. If the voltage starts to rise on the input supply continue increasing the input supply current limit while watching the output voltage. If the current increases on the input supply, but the voltage remains near zero, then there may be a short or a component misplaced on the board. Power down the board and visually inspect for solder bridges and recheck the diode and capacitor polarities. Once the power supply circuit is operational then more extensive testing may include full load testing, transient load and line tests to compare with simulation results.

Load Testing

The setup is the same as the initial startup, except that an additional digital voltmeter is connected between Vin and GND, a load is connected between Vout and GND and a current meter is connected in series between Vout and the load. The load must be able to handle at least rated output power + 50% (7.5 watts for this design). Ideally the load is supplied in the form of a variable load test unit. It can also be done in the form of suitably large power resistors. When using an oscilloscope to measure waveforms on the prototype board, the ground leads of the oscilloscope probes should be as short as possible and the area of the loop formed by the ground lead should be kept to a minimum. This will help reduce ground lead inductance and eliminate EMI noise that is not actually present in the circuit.



Design Assistance

1. Master key : 84053E93FED573F79AD11CC824948899[v1]

2. **TPS40211** Product Folder : http://www.ti.com/product/TPS40211 : contains the data sheet and other resources.